Portfolio Stress Testing & Monte Carlo Simulation

Overview

This project develops a financial risk analysis pipeline to evaluate how a multi-asset portfolio behaves under both normal and extreme market conditions.

It combines:

- **Historical risk analysis**
- **Scenario-based stress testing**
- **Monte Carlo simulation**

The objective is to quantify downside risk, tail exposure, and portfolio vulnerability using real market data.

Tools & Technologies

- **R (dplyr, ggplot2, MASS, tidyr)**
- **MySQL (data storage)**
- **Yahoo Finance (data source)**

Portfolio Composition

- **Apple stock (AAPL) – Large-cap technology**
- **JP Morgan Stock (JPM) – Banking sector**
- **Tesla Stock (TSLA) – High-growth / high-volatility**
- **Vanguard FTSE EFT Stock (VWO) – Emerging markets ETF**

Equal-weighted portfolio (25% each)

Methodology

1. Data Pipeline

- **Data extraction via Yahoo Finance API**
- **Storage in MySQL database**
- **Retrieval and transformation in R**

2. Portfolio Construction

- **Log returns computed for each asset**
- **Portfolio returns calculated using equal weights**

3. Risk Metrics

- **Value at Risk (VaR)**
- **Expected Shortfall (ES)**
- **Volatility and distribution analysis**

4. Stress Testing Scenarios

- **Market Crash (-20% shock across assets)**
- **Interest Rate Shock (sector-specific impact)**
- **Emerging Market Crisis (localized shock to VWO)**

5. Monte Carlo Simulation

- **10,000 simulated scenarios**
- **Multivariate normal distribution**
- **Portfolio return distribution analysis**

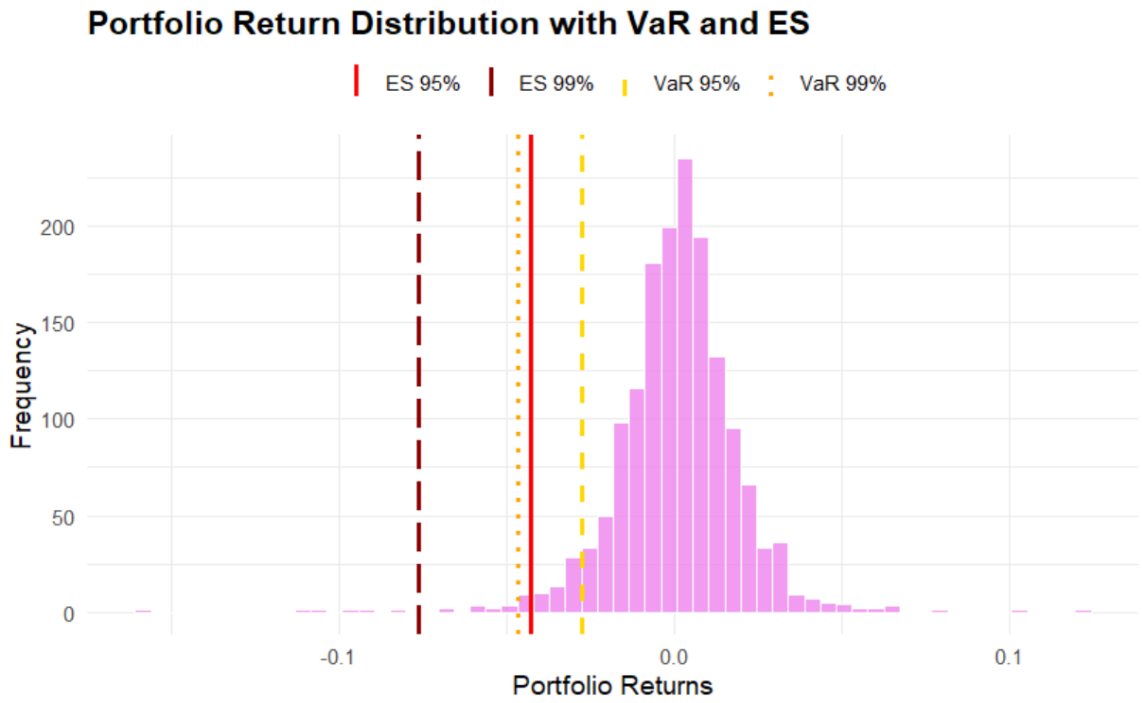
Key Observations

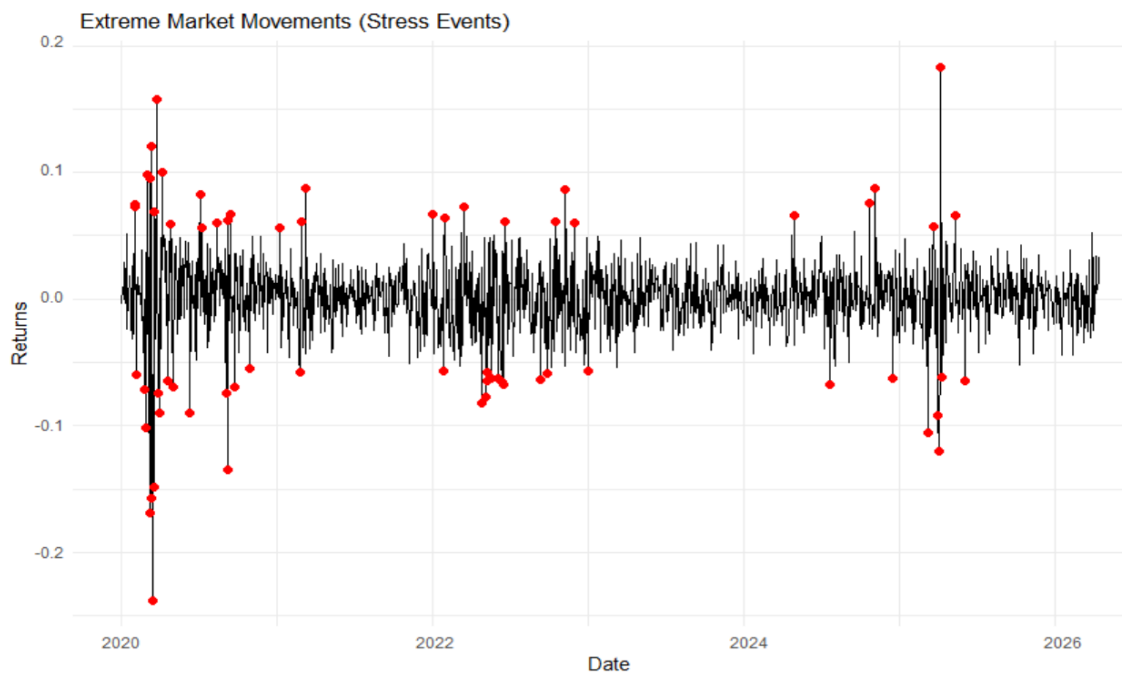
- **Portfolio risk is non-linear under stress**
- **Correlations increase during crises, reducing diversification**
- **Tail risk is substantial and cannot be captured by volatility alone**
- **Historical models underestimate extreme losses**

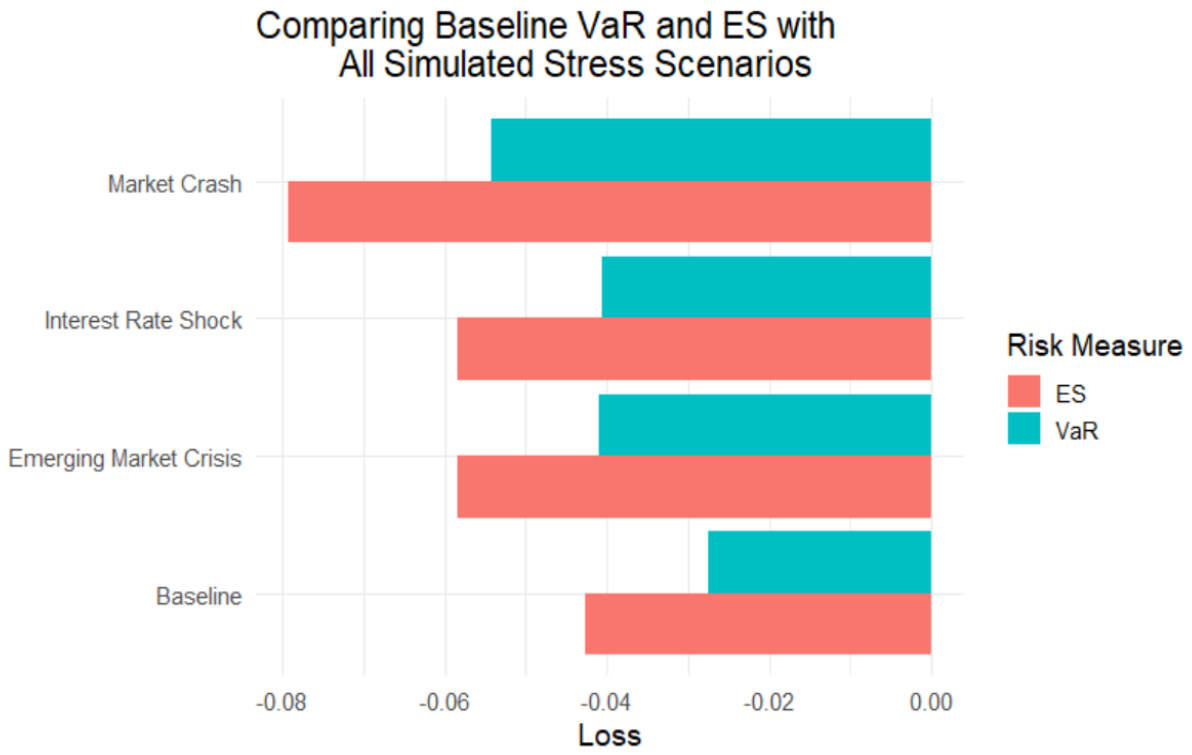
Key Results

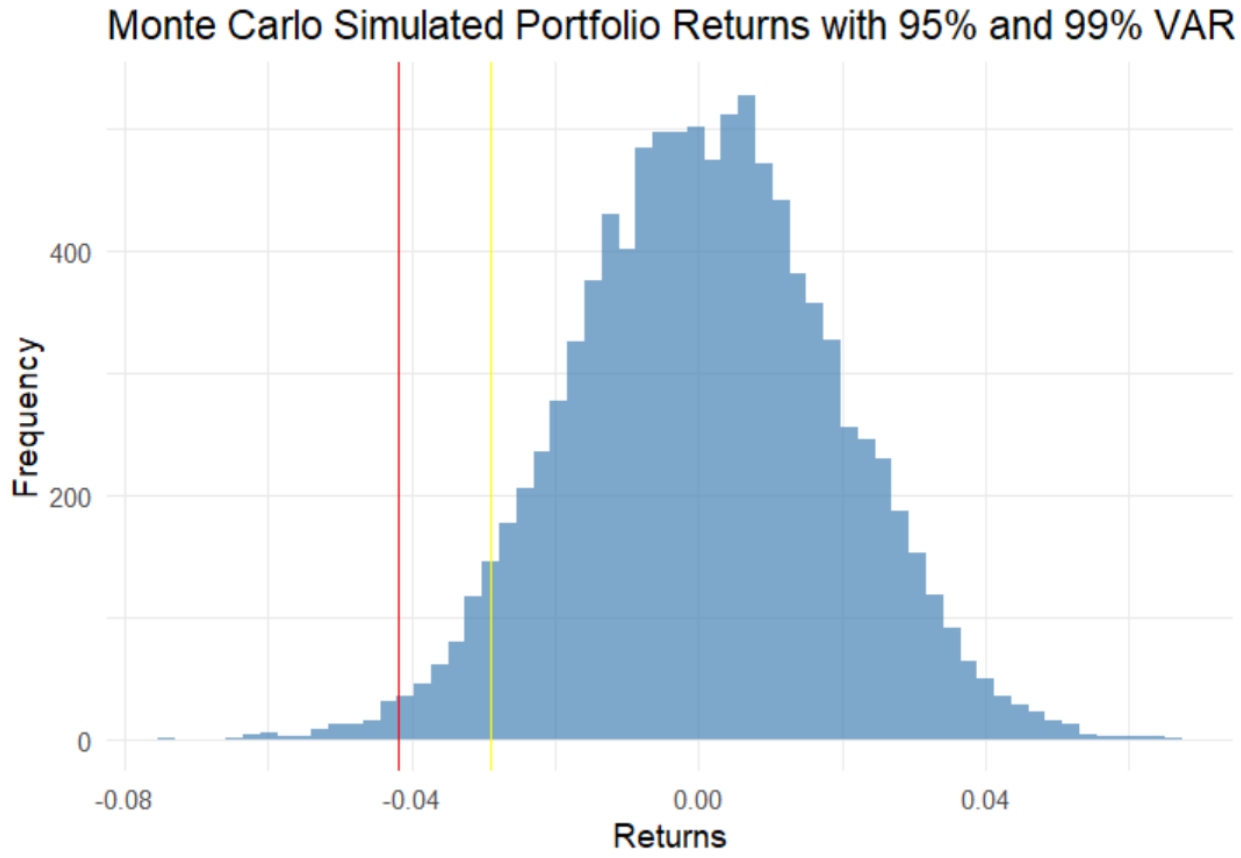
- **Stress scenarios significantly increase downside risk**
- **Market crash produces the most severe losses**
- **Diversification benefits weaken during systemic shocks**
- **Expected Shortfall consistently exceeds VaR, highlighting tail risk**
- **Monte Carlo simulation confirms risk but may underestimate extreme events**

Sample Outputs






Comparing Baseline VaR and ES with
All Simulated Stress Scenarios
Market Crash
Interest Rate Shock
Emerging Market Crisis
Baseline
-0.08
-0.06
-0.04
-0.02
0.00
Loss
Risk Measure
ES
VaR


Monte Carlo Simulated Portfolio Returns with 95% and 99% VAR
Frequency
400
200
0
-0.08
-0.04
0.00
0.04
Returns

How to Run

1. **Clone the repository**
2. **Open R project**
3. **Run scripts in order:**

    Script 1 data extraction and sql connection.R

    Script 2 stock return computation and exploratory data analysis.R

    Script 3 portfolio construction and risk visualization.R

    Script 4 scenario analysis and stress testing.R

Skills Demonstrated

- **Financial risk modeling**
- **Data engineering (SQL + R pipeline)**
- **Statistical analysis**
- **Monte Carlo simulation**
- **Scenario-based stress testing**

Future Improvements

- **Using fat-tailed distributions (t-distribution)**
- **Employing volatility modeling (GARCH)**
- **Dynamic portfolio optimization**
- **Macro-driven stress scenarios**

<u>Author</u>

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